Anagol et al. (2024) Diffuse Bunching with Frictions

A Reading Guide to the Sparsity-Based Estimator

Zhongji Wei

April 27, 2026

Outline

1. Background: from Saez to diffuse bunching
2. Anagol's method: a sparsity model
3. Where the framework is — and is not — applicable

Why this paper?

- Many bunching datasets show:
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  - A positive density inside the dominated region (above a notch)
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- **Anagol et al. (2024)** proposes a **three-parameter** sparsity-based framework that captures all of these features in one parsimonious model.
- Today: (i) walk through the math; (ii) understand what Figure 10 (VAT registration threshold) achieves; (iii) discuss when this estimator is the right tool.

Saez (2010) frictionless baseline

In a frictionless static model, a tax kink/notch at k produces an **atom of mass**:

$$B \;=\; H_1(k) - H_0(k) \;=\; \int_{-\infty}^{k} (h - h_0)\, dz \;+\; \int_{k}^{\infty} (h - h_1)\, dz$$

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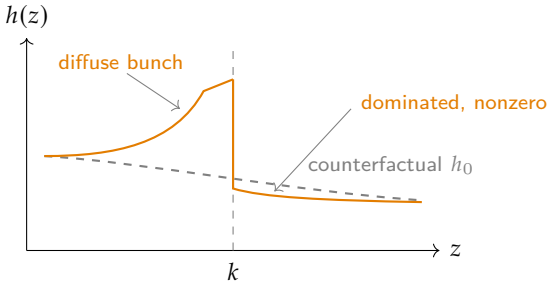
Taylor-expanding H_0 at $z = k$ rewrites B as a function of the same-quantile horizontal shift Δz (i.e., $H_0(k + \Delta z) = H_1(k)$):

$$B \;=\; h_0(k)\,\Delta z \;+\; \frac{h_0'(k)}{2}\,\Delta z^2 \;+\; \cdots$$

Under iso-elastic utility, the standard Saez conversion gives the elasticity:

$$e \;\approx\; \frac{\Delta z / k}{\ln\big((1 - t_0)/(1 - t_1)\big)}$$

What we observe is not an atom — it is a diffuse mass



(illustrative; not real data)

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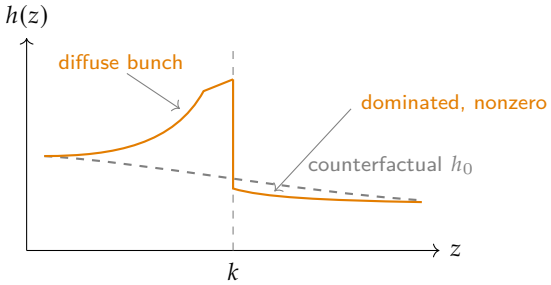

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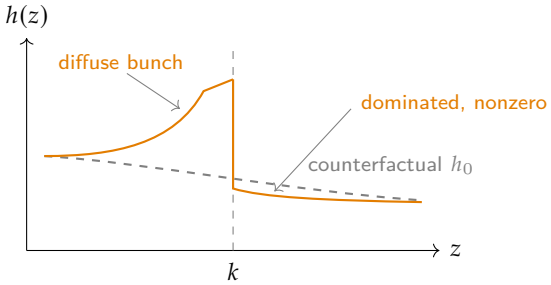

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Anagol's move: model friction explicitly as an endogenous output.

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Anagol modification (§2.2): each taxpayer picks the highest-utility option from a **discrete opportunity set**:

$$\left\{ z^*(n) + \varepsilon_1,\ \ z^*(n) + \varepsilon_2,\ \ \ldots,\ \ z^*(n) + \varepsilon_M \right\}$$

- $z^*(n)$: type-n frictionless preferred income (the "target")
- ε_i are i.i.d. within type, with distribution $F_\varepsilon(\cdot|n)$ and density $f_\varepsilon(\cdot|n)$
- The pair (F_ε, M) fully characterizes the friction process

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Microfoundations compatible with this object: search costs, lumpy adjustment, rational inattention, misperception. **Key idea**: Anagol does not commit to a particular microfoundation — instead, he proves a universality result that bypasses the choice.

Proposition 1: the type-conditional density

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For type n with utility $u(c, z|n)$ and tax schedule $T(z)$, the type-conditional income density at $\bar{z}$ is

$$g(\bar{z}|n) \;=\; \underbrace{M \cdot f_\varepsilon\big(\bar{z} - z^*(n)\,\big|\,n\big)}_{\bar{z}\text{ enters the set}} \times \left[1 - \int_{\Theta(\bar{z}|n)} f_\varepsilon\big(z - z^*(n)\,\big|\,n\big)\,dz\right]^{M-1}$$

where $\Theta(\bar{z}|n) = \{\, z : u(z - T(z), z|n) \geq u(\bar{z} - T(\bar{z}), \bar{z}|n) \,\}$ is the *utility-dominating region*.

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Two-step intuition (the product structure):

1. $M \cdot f_\varepsilon$: the (density of the) probability that $\bar{z}$ shows up in the opportunity set
2. $[\cdot]^{M-1}$: the probability that none of the remaining $M - 1$ draws lies in Θ — i.e., no better alternative steals the agent away from $\bar{z}$

Linear tax + concave indirect utility $\Rightarrow$ Θ is a single interval; under a notch Θ may be a union of two disjoint

Proposition 3: Universality —— frictions converge to a one-parameter limit

Prop 1 is elegant but not parsimonious: to fit data, one must specify (F_ε, M).
The leap: as $M \to \infty$ with F_ε horizontally compressed by $1/M$, the model converges to a **one-parameter** uniform sparsity model (the opportunity set becomes a Poisson process).

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Suppose F_ε is C^1 at 0 with $f_\varepsilon(0) > 0$. Define the rescaled family $F_\varepsilon^M(x) := F_\varepsilon(x/M)$ (so $M \cdot f_\varepsilon^M(0) = \lambda$ is constant). As $M \to \infty$, the type-conditional density converges pointwise to

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Implication: the fitting layer reduces to a **single parameter** λ, or equivalently $\mu := 1/\lambda$, called *lumpiness* (the average distance between adjacent opportunities). $\mu \downarrow 0$ recovers the frictionless model; larger μ produces more diffuse bunching.

Proposition 3: proof sketch

Write the boundary of $\Theta(\bar{z}|n)$ in error space as $[\underline{\phi}(x), \overline{\phi}(x)]$ where $x = \bar{z} - z^*(n)$. From Prop 1:

$$g^M(z^* + x \,|\, n) = f_\varepsilon\left(\frac{x}{M}\right)\left[1 - \left(F_\varepsilon\left(\frac{\overline{\phi}(x)}{M}\right) - F_\varepsilon\left(\frac{\underline{\phi}(x)}{M}\right)\right)\right]^{M-1}$$

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Takeaway: the choice of microfoundation (search cost / inattention / misperception) is bypassed; *any* friction satisfying the C^1 regularity condition is approximated by the same one-parameter limit model.

Proposition 4: quadratic approximation ⟹ Laplace form

Prop 3 still requires computing $|\Theta(\bar{z}|n)|$ — generally numerical.

Assumption (premise of Prop 4)

$v(z|n) \approx v(z^*(n)|n) + \frac{1}{2}\, v''(z^*(n)|n)\,(z - z^*(n))^2$ (first-order term vanishes at the optimum).

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Why elegant: g does not depend on the elasticity e or curvature v'' directly. v'' symmetrically determines both endpoints of Θ, but the length $|\Theta|$ depends only on the distance $|\bar{z} - z^*|$ — v'' cancels.

Corrections to Θ under kinks and notches

Proposition 4 is stated for a linear tax (no kink, no notch).

Kink: indirect utility is no longer globally quadratic, but it is *piecewise* quadratic (T_0 branch and T_1 branch); Θ's endpoints are solved branch by branch. Under the local quadratic approximation, the boundaries are still roots of **quadratic equations** — closed-form.

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Implication

A single estimator handles kinks and notches uniformly — only the form of Θ changes, not the estimator's logic.

Estimator: polynomial type density + MLE

The observed density is the type-conditional density integrated against the type distribution:

$$h(z) = \int_n g(z|n)\, f(n)\, dn$$

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- the polynomial is fit to the *type distribution* $f(n)$, not to the observed density $h(z)$
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Likelihood: $L(e, \mu, dT, \theta) = \prod_i h(X_i|\cdot)$. Linearity of $h(z)$ in θ gives a closed-form inner solution for the polynomial; only (e, μ, dT) are maximized in the outer loop.

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Lemma 1 — three-piece decomposition of the threshold density

Under the uniform sparsity model:

$$h(k) = \underbrace{\lambda B}_{\text{frictionless excess}} + \underbrace{\int_{-\infty}^{k} h_0^*(z_0^*)\,\lambda\, e^{-2\lambda|k-z_0^*|}\,dz_0^*}_{\text{contribution of below-threshold types}} + \underbrace{\int_{k}^{\infty} h_1^*(z_1^*)\,\lambda\, e^{-2\lambda|k-z_1^*|}\,dz_1^*}_{\text{contribution of above-threshold types}}$$

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Identification (kink, two-step): far-tail density pins $B \equiv \Delta z$ (standard Saez); given Δz, the shoulder decay identifies λ (equivalently $\mu = 1/\lambda$). Relies on polynomial target density. Notches with unknown dT: joint MLE; Anagol App. D shows 95% CI of $(\hat{e}, \hat{\mu}, \hat{dT})$ covers truth.

Notch: a non-trivial phenomenon — double local maxima

Notch vs. kink: under a notch, $T_1(z)$ creates a **discontinuity** above the threshold.

The marginal type c's indirect utility has **two local maxima**:

- one just below the threshold k
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Implication

The type-conditional density can be **multimodal**: aside from the main bunch, there is a non-zero probability of "jumping over the threshold to the far peak", producing a **secondary peak** in the data.

Figure 10: VAT registration threshold, three-parameter fit

Setting: South Africa, turnover at 1 million ZAR is the mandatory VAT registration threshold; registration imposes a compliance cost.

Observed density (green dots):

- Below 1M: gentle baseline → sharp rise to peak ≈ 8 thousand firms/bin — **diffuse left bunch**
- Just above 1M: vertical drop to ≈ 2.3 (not zero)
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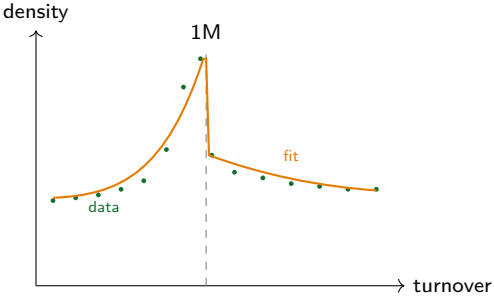
Three-parameter fit (orange line):

- $\hat{e} = 0.54$ (taxable-turnover elasticity)
- $\hat{\mu} = 69.9$k ZAR (avg. distance between opportunities)
- $\hat{dT} = 6.47$k ZAR turnover "as-if notch" → via 6.18% avg. profit rate → **R399.846 compliance cost** for threshold optimization



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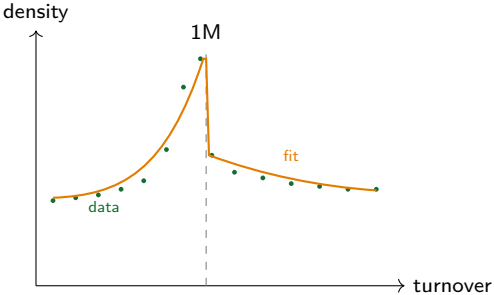

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Punchline: both the diffuse bunch and the dominated-region density are captured by the same three parameters, with **no excluded window** required.

Implicit assumptions one should keep in mind

Before applying Anagol's estimator, it is worth surfacing the assumptions:

1. $f_\varepsilon(0) > 0$ **and** $F_\varepsilon \in C^1$ **at** 0 (Prop 3 universality premise)
2. **Indirect utility is locally quadratic at the frictionless optimum** z^* (Prop 4 Laplace form premise; non-integer power utilities break this)
3. **The type distribution** $f(n)$ **is well-approximated by a low-order polynomial near the threshold** (the core working assumption for the estimator)
4. **The target-income counterfactual densities** h_0^*, h_1^* **are smooth near the threshold** (implicit identification condition behind Lemma 1)
5. Θ **is determined purely by static utility comparison** (no dynamic / strategic interaction)
6. **Reporting noise, if absorbed into** μ**, is an identification simplification** — not a theoretical equivalence (normal smoothing $\neq$ sparsity diffusion)
7. **Observations are i.i.d. for MLE**

Summary

- Anagol et al. (2024) replaces the implicit-friction tradition (Chetty / Kleven–Waseem) with an explicit **sparsity model**: agents choose from a discrete opportunity set, not a continuum.
- A universality result (Prop 3) collapses every micro-friction satisfying mild regularity into a **one-parameter** uniform sparsity limit, parameterized by lumpiness μ.
- A quadratic approximation (Prop 4) yields a **Laplace-form** type-conditional density; kinks and notches are handled by branch-wise quadratic equations on Θ.
- Identification (Lemma 1) separates the elasticity e from the lumpiness μ: the threshold peak height pins Δz; the shoulder decay pins λ.
- The estimator captures both the diffuse bunch and the positive dominated-region density with three parameters (e, μ, dT) — no excluded window.
- Notches generate **multimodal** type-conditional densities, a testable signal whenever a secondary peak appears above the threshold.

Discussion prompts

- **Scope**: For which datasets is the sparsity framework strictly preferable to Chetty / Kleven–Waseem? When does the data not warrant the extra structure?
- **Microfoundations**: Which microfoundation (search costs / inattention / lumpy adjustment) maps to which range of μ? Is μ economically meaningful, or is it a reduced-form summary?
- **Robustness of Prop 4**: How sharply does the Laplace-form approximation deteriorate when indirect utility deviates from local quadratic? (E.g., non-integer power utilities, regime-changing audit costs.)
- **Multimodal data**: When secondary peaks are observed above a notch, what additional models (audit probability, dynamic strategic interaction) become candidates?
- **Welfare**: The estimator delivers $(\hat{e}, \hat{\mu}, \widehat{dT})$ but not welfare directly. What additional structure is needed to translate the sparsity friction into deadweight loss?

Thank you!

Comments and questions welcome.

Appendix: Θ under a notch — details

Under a linear tax with concave indirect utility, $\Theta(\bar{z}|n)$ is a **single interval** $[\underline{Z}, \overline{Z}]$.

Under a notch: T_1 has a discontinuity above the threshold; indirect utility develops two local maxima:

- one just below the threshold
- the other at $z_1^*(c) > k$ in the far region

Θ becomes a union of two disjoint intervals:

$$|\Theta(\bar{z}|c)| = \underbrace{(\overline{Z}_1 - \overline{Z}_0)}_{\text{main interval}} - \underbrace{(\overline{Z}_1^{\text{lower}} - k)}_{\text{dominated band}}$$

The type-conditional density is multimodal: aside from the main bunch, a secondary peak emerges above the threshold (cf. Anagol Figure A2 (d) for the two-peak indirect utility, Figure A3 (b) for the multi-modal density).

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Appendix: Prop 4 curvature decoupling — derivation

Under the quadratic-utility assumption $v(z|n) = v(z^*) + \frac{1}{2}v''(z - z^*)^2$ with $v'' < 0$:

The condition $v(z|n) \geq v(\bar{z}|n)$ becomes

$$\tfrac{1}{2}v''(z - z^*)^2 \;\geq\; \tfrac{1}{2}v''(\bar{z} - z^*)^2 \quad\Longleftrightarrow\quad (z - z^*)^2 \;\leq\; (\bar{z} - z^*)^2 \quad\Longleftrightarrow\quad |z - z^*| \;\leq\; |\bar{z} - z^*|$$

Hence $\Theta(\bar{z}|n) = [\, z^* - |\bar{z} - z^*|,\; z^* + |\bar{z} - z^*| \,]$ and

$$|\Theta(\bar{z}|n)| \;=\; 2\,|\bar{z} - z^*|$$

Crucially: v'' enters both endpoints symmetrically; the length $|\Theta| = 2|\bar{z} - z^*|$ depends only on the distance $|\bar{z} - z^*|$, with v'' canceling. The lumpiness limit density therefore depends only on the location z^* and the parameter λ.

Caveat: this decoupling is local. Far from z^*, or under a notch, the quadratic approximation breaks down and $|\Theta|$ regains a curvature dependence — see Prop 4's branch-wise treatment.

Appendix: identification of dT for notches

Lemma 1 is stated for a kink with known counterfactuals. For a notch with unknown effective dT, separate identification of (e, μ, dT) is not given by a closed-form proof — Anagol's App. D supplies simulation-based evidence.

Identification signals in the data:

- Total bunching mass $\longrightarrow$ pins e (via Δz, Saez)
- Shoulder decay rate near the threshold $\longrightarrow$ pins λ
- Width of the dominated band on the high side $\longrightarrow$ pins dT
- Secondary peak (if present) above the threshold $\longrightarrow$ informs the curvature of indirect utility, supplemental to dT

Simulation result: 95% confidence intervals for $(\hat{e}, \hat{\mu}, \hat{dT})$ cover the true values across a wide range of (e, μ, dT) settings. The joint distribution shows a positive correlation between $\hat{e}$ and $\hat{\mu}$, but the parameters are not collinear.